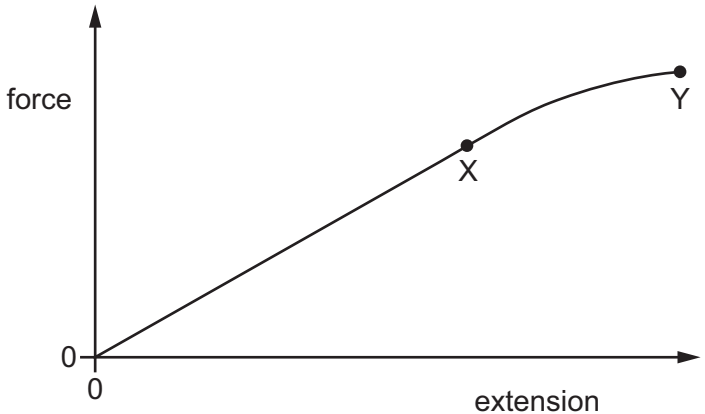
**18** A sample of metal is subjected to a force which increases to a maximum value and then decreases back to zero. A force–extension graph for the sample is shown.




When the sample contracts, it follows the same force–extension curve as when it was being stretched.

What is the behaviour of the metal between X and Y?

**A** both elastic and plastic

**B** **not** elastic and **not** plastic

**C** elastic but **not** plastic

**D** plastic but **not** elastic